Lecture - 2

❖ Evolution of Programming Languages

Programming languages have evolved from **machine-oriented languages** to **human-friendly and problem-oriented languages**.

- **First Generation Language (1GL) – Machine Language**
  - Machine language is the lowest-level programming language.
  - Instructions are written using 0s and 1s (binary).
  - It can be directly understood and executed by the CPU.
  - It is very fast but extremely difficult for humans to write, understand, and debug.
  - It is machine dependent.

    Example:

    10110000 01100001
- **Second Generation Language (2GL) – Assembly Language**
  - Assembly language uses short words called mnemonics instead of binary.
  - Examples of mnemonics: MOV, ADD, SUB, JMP.
  - It is easier to understand than machine language.
  - An assembler converts assembly language into machine code.
  - It is generally hardware/machine dependent.

    Example:

    ```
    MOV A, B
    ADD A, C
    ```
- **Third Generation Language (3GL) – High-Level Language**
  - High-level languages are designed to be easy for humans to read, write, and understand.
  - They use English-like statements and mathematical expressions.
  - They are generally more portable than assembly language.
  - A compiler or interpreter translates the program into a form the computer can execute.

    Examples: C, C++, Java, Python, FORTRAN, Pascal.

    Example in C:

    ```
    sum = a + b;
    ```
- **Fourth Generation Language (4GL)**
  - 4GL provides a higher level of abstraction than 3GL.
  - It requires fewer lines of code to perform tasks.
  - It is often used for database operations, report generation, and data analysis.
  - The programmer specifies what is required, rather than describing every step to achieve it.
  - SQL is a common example.

    Example:

    ```
    SELECT name, marks FROM student WHERE marks > 80;
    ```
- **Fifth Generation Language (5GL)**
  - 5GL is mainly associated with Artificial Intelligence (AI) and logic/constraint-based programming.

- The programmer specifies the problem, rules, constraints, or desired result.
- The system determines how to arrive at the solution.
- Prolog is a classic example.
- It is used in areas such as AI, expert systems, and knowledge-based applications.
  Example idea:
  Facts + Rules + Query → Solution

Comparison Table

Generation	Language	Main Idea	Example
1GL	Machine Language	0s and 1s	10110000
2GL	Assembly Language	Mnemonics	MOV, ADD
3GL	High-Level Language	Human-readable programming	C, Java, Python
4GL	Very High-Level Language	Specify **what** is needed	SQL
5GL	AI/Logic Language	Specify problem, rules, constraints	Prolog

❖ **Language Translators**

These are the programs which are used for converting the programs in one language into Machine language instructions, so that they can be executed by the computer.

1. **Assembler:** It is a program which is used to convert the assembly level language programs into machine language.
2. **Compiler:** It is a program which is used to convert the high level language programs into machine language.
3. **Interpreter:** It is a program, it takes one statement of a high level language program, translates it into machine language instruction and then immediately executes the resulting machine language instruction and so on.

<u>Sample Important Questions:</u>

1. Explain the evolution of programming languages from 1GL to 5GL with examples.
2. Compare 1GL, 2GL, 3GL, 4GL and 5GL.
3. Explain assembler, compiler and interpreter.
4. Differentiate between compiler and interpreter.
5. Explain the features and limitations of Machine Language and Assembly Language.